

1. What is the objective of the generator in the original GAN formulation?

- a. Minimize the discriminator's accuracy on real data
- b. Maximize $\log(Dx)$
- c. Minimize $\log(1 - D(G(z)))$
- d. Reduce the number of generated samples

Answer: C

2. Why does the original GAN generator suffer from the issue of vanishing gradients?

- a. The generator has too many parameters
- b. The discriminator becomes too weak
- c. The discriminator becomes too strong and outputs values close to 0 or 1
- d. The learning rate is too high

Answer: C

3. If the discriminator outputs $D(G(z)) = 0.1$, what does the generator try to do?

- a. Decrease it further toward 0
- b. Increase it toward 1
- c. Keep it unchanged
- d. Minimize $D(x)$

Answer: B

4. The Original GAN objective is:

$$\min_G \max_D V(D, G)$$

$$\text{Where: } V(D, G) = E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p_z} [\log (1 - D(G(z)))]$$

What does this formulation represent?

- a. A supervised learning problem
- b. A minimization problem for both networks
- c. A clustering objective
- d. A competitive game between generator and discriminator

Answer: D

5. Suppose the discriminator outputs:

$$D(x) = 0.9 \text{ for a real image}$$

$$D(G(z)) = 0.2 \text{ for a fake image}$$

What is the discriminator loss assuming

$$L_d = -[\log(D(x)) + \log(1 - D(G(z)))]$$

- a. $-\log(0.9) + \log(0.8)$
- b. $\log(0.9) - \log(0.2)$
- c. $-\log(0.1) + \log(0.8)$
- d. $\log(0.1) - \log(0.2)$

Answer: A